Sigmawave

Computer Vision Assessment Task: Black Gun Detection v1.0.2

Objective

To evaluate the candidate's ability to prepare, train, and evaluate a computer vision model for detecting a **Black Gun (KAC PDW)** using both

Synthetic Data (SD) : “synthetic_dataset_KAC_PDW_Blackgun_v2”

Real Data : “real_dataset_KAC_PDW_Blackgun”

within the TRMS system, and to compare their performance.

Duration: 1 week

Assessment Deliverables

1. **Data Preparation**
    a. Organize datasets for:
        - VSD (Synthetic Data): Pre-generated or provided.
        - Real Data: Captured from actual camera/test footage.
    b. Annotate or validate existing annotations (bounding boxes or segmentation).
    c. Create a clear structure for training and testing.
2. **Model Training**
    a. Train a computer vision model (e.g. YOLO) using
        - Synthetic Data only
        - Real World Data Only
        - Combine Synthetic Data with Real World Data
3. **Test Video Inference**
    a. Run inference on test videos captured in real environments.
    b. Save inference outputs (frames or videos with detected bounding boxes).
4. **Reporting**
    a. Include the following in your report:
        - Detection Accuracy
        - Qualitative Examples (screenshots from test videos)
        - Failure Cases
        - Comparison of SD-only model vs Real-data-trained model
        - Performance differences on Real Test Video
        - Lighting conditions (Day / Night mode)
    b. Optionally, suggest how to better structure or capture future test video data to improve performance

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   c. Provide detailed analysis on the real vs synthetic datasets and highlight or comment on the accuracy, precision and important statistics that might convince people to use or not to use synthetic datasets instead of real data sets.
   d. Our business is in Visual Synthetic Data, we would like to know this quality of the data sets and if they can be used for real world or not. If yes or no, why?
5. BONUS: Implement simple object tracking so the gun is assigned an ID (e.g., "Weapon ID 01") across frames.
6. Additional: To show robustness of solutions, candidates are to ensure the model is robust enough to do a detection on a video release 2h before the actual review interview, and to produce results and table reporting as the other test videos. This will be released on the day of the review interview itself.
7. Ensure total data isolation by excluding your test videos from the training and validation sets. These videos must serve as an independent Ground Truth benchmark. You are required to manually annotate these test frames to calculate a Confusion Matrix (identifying True Positives, True Negatives, False Positives, and False Negatives). Use these counts to derive standard performance metrics, including Accuracy, Precision, and Recall, to evaluate the model's real-world reliability.

Tools & Constraints

- Annotation Format & Submission Format: YOLO format
- Submission Format:
  - Python scripts or notebooks
  - Inference result videos/images
  - Report (PDF or Markdown)
  - Submission must be pushed to a public or private Git repository (e.g., GitHub, GitLab, Bitbucket), include read.me of how to run the code.
- You are allowed to do your own additional annotations and labellings on the datasets for more accuracy and precision.

Bonus

A new set of cleaner VSD_v3, is available as “synthetic_dataset_KAC_PDW_Blackgun_v3”

Do a comparison with

- Real World Footage
- VSD v2: real_dataset_KAC_PDW_Blackgun_v2

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